School of Computer Engineering

Manipal Institute of Technology

Manipal Academy of Higher Education

Manipal, Karnataka, India

Auditing the Faithfulness and Cross-City Consistency of Attention in Multimodal Spatiotemporal Air Quality Index Prediction

Methodology Submission

ML Lab Project - 10 October 2026

Including same-day estimation and next-day forecasting

Team member	Registration number	Roll number
Saanvi Garg	240905702	68
Sparsh Agrawal	240905708	70
Dhruv Chamria	240905706	69
Neelaksha Sisodiya	240905642	65

Section CSE-C

Faculty guide: Dr Nisha P Shetty

Academic year: 2026-2027

A. Overall System Architecture

A.1 Research objective and prediction tasks

The project studies air quality index (AQI) prediction across Indian monitoring stations and examines whether learned attention agrees with the inputs that influence a trained model. The implemented system has two linked components: an exploratory multimodal attention study, and chronological benchmarks for practical AQI estimation and forecasting. Prediction accuracy and explanation faithfulness are evaluated separately; attention weights are treated as hypotheses about model behaviour rather than causal explanations [1].

The prediction unit is a monitoring station s on a target date d. The response $y_{s,d}$ is the quality-filtered reported end-of-day AQI. In **same-day estimation**, the model uses environmental observations for date d together with calendar, location and eligible historical AQI features. This is a retrospective estimation task because the day's satellite products and reanalysis weather may become available after the nominal prediction time.

In **next-day forecasting**, the model predicts date d using AQI history ending on d - 1. The operational sequence model uses 28 preceding calendar days. A separate weather-aware candidate additionally uses archived forecasts made before the target-day boundary. Target-day observed AQI, satellite retrievals and realized weather are excluded from this forecasting branch.

A.2 End-to-end workflow

Inputs Reported station AQI; Sentinel-5P NO2/HCHO; MODIS AOD; ERA5-Land weather; archived GFS weather forecasts; station coordinates
Data preparation Timestamp conversion, duplicate and quality checks, station-date target construction, spatial joins, exact calendar lags and missingness indicators
Experiment construction Training-only preprocessing; chronological fitting, selection, calibration and held-out testing; separate attention/control experiments
Outputs and audit Station predictions, city summaries, calibration intervals, baseline comparisons, attention perturbation audits and downloadable dashboard results

Fig. 1 System workflow. Observed target-day environmental inputs belong only to the retrospective estimation and research branches.

A.3 Scope of interpretation

The study separates improved prediction from improved explanation. The deployed Ridge, plain GRU and weather-aware XGBoost models do not expose learned attention maps. Attention conclusions therefore refer to the dedicated research models, while forecasting conclusions refer to independent future-year benchmarks. Historical input cutoffs are enforced, but source publication latency has not been independently reconstructed for every observation.

B. Dataset Description

B.1 Sources, geographical scope and sample definition

The station configuration covers **107 monitoring stations in eight cities**: Bengaluru, Chennai, Delhi, Hyderabad, Jaipur, Kolkata, Lucknow and Mumbai. Reported AQI is collected from the CPCB repository. Environmental extraction covers 2023-2025; the subsequent forecast test uses available reported AQI from January-September 2026. Station coordinates and city membership come from the project configuration.

Table 1 Data sources and their roles. Satellite columns and AOD are not direct ground-level pollutant concentrations.

Source	Variables and implemented processing	Role
CPCB repository	Hourly reported AQI; station and date identifiers; daily quality filters	Primary response and lagged history
Sentinel-5P OFFL L3	Tropospheric NO2 column and HCHO column, aggregated over daily extraction windows	Satellite inputs
MODIS MAIAC	MCD19A2.061 Optical_Depth_047; scale factor 0.001 [5]	Aerosol input
ERA5-Land DAILY_AGGR	Temperature, dew point, surface pressure, precipitation, 10 m u/v wind [6]	Retrospective weather
Open-Meteo previous runs	GFS Seamless previous_day1 hourly forecasts; complete daily aggregation [7]	Forecast-safe weather candidate

B.2 Response definition and quality rules

Raw timestamps are interpreted in Indian Standard Time (IST) and converted to UTC before date alignment. The implemented AQI_eod endpoint is the reported reading at **05:00 IST on the following morning**, corresponding to 23:30 UTC on date d. It serves as the project's daily endpoint approximation. This response is already reported AQI; it is not a new AQI reconstructed from a subset of pollutant concentrations.

$$y_{s,d} = \text{reported AQI}_s(\text{23:30 UTC on date d}) \quad (1)$$

An eligible station-day requires at least 12 valid hourly readings and a non-missing endpoint. Values outside 0-500 are rejected. Constant hourly series with equal daily minimum and maximum below 500 are flagged as stuck and excluded. Identical duplicate records are removed; conflicting duplicates are checked by the multiyear preparation pipeline. Missing targets are never imputed.

The endpoint can require a file from the following local calendar month. Missing boundary readings therefore remain missing. Daily mean AQI is retained as an auxiliary summary and is not substituted for the primary response. Station export counts, usable station-day counts and final paired evaluation counts are reported separately.

B. Dataset Description (continued)

B.3 Spatial alignment, feature construction and missingness

Google Earth Engine exports are sampled on city grids spaced by 0.03 degrees, approximately 3 km. Station records are joined to the nearest grid point. A nominal sampling scale of 1,000 m is used, but this does not increase the native resolution of a coarse weather product. Satellite gaps are preserved as missing values; an empty satellite collection does not cause an otherwise usable weather day to be discarded.

Table 2 Implemented feature groups. Feature availability depends on the experiment and prediction mode.

Feature group	Variables / construction
Satellite (S)	NO2 column, HCHO column and MODIS AOD at 0.47 micrometres
Weather (W)	Temperature in Celsius; relative humidity; wind speed; sine/cosine of wind direction; pressure in kPa; precipitation in mm
Geography (G)	Station latitude and longitude
Time and weather history (T)	Sine/cosine of day of year; weather lags of 1 and 2 days; preceding 3-day wind mean and rainfall sum
AQI history (H)	AQI at d - 1, d - 2 and d - 7; preceding 3-day AQI mean
Long forecast history	28 calendar-day AQI lags; count, mean and standard deviation over 3, 7, 14 and 28 days; latest AQI change
Archived forecast weather	Daily temperature, relative humidity, wind speed, pressure and total precipitation

Temperature is converted from kelvin to Celsius. Relative humidity is derived from temperature and dew point and constrained to 0-100%. Wind speed is the magnitude of the u/v vector, and circular wind direction is represented by sine and cosine. Pressure is converted from Pa to kPa and precipitation from metres to millimetres, with negative precipitation constrained to zero.

Lag construction uses exact station-date lookup or a reindexed daily calendar. A missing calendar day remains a missing lag; the previous available row is never treated as the previous day. Tabular pipelines use training-fitted median imputation and missingness indicators where configured. Sequence inputs retain an explicit AQI-availability channel and fill missing standardized numerical inputs with zero, corresponding to the fitted mean.

B.4 Archived forecast issue-time control

The weather candidate uses GFS Seamless archived previous-day forecasts at city-level coordinates formed from the mean of configured station locations. Each target-valid hour uses a forecast made 24 hours earlier. All included issue times precede the target-day 00:00 UTC boundary, but the daily feature combines multiple hour-specific earlier runs rather than one common issuance. All 24 hourly values must be present for each daily variable. Missing forecast values are not replaced by realized target-day weather.

This archive supports an input-cutoff-controlled historical evaluation. It does not by itself verify real-time ingestion delays or fine-scale weather at every station. The 2026 benchmark does not fabricate observed satellite or ERA5 inputs for that year.

C. Proposed Model / Algorithm

C.1 Baselines and same-day estimation

Persistence predicts the target-day AQI as the immediately preceding calendar-day AQI. It is evaluated on the same eligible rows as each competing model. This baseline is essential because strong autocorrelation can make a shifted observed curve appear visually convincing without adding forecast skill.

$$\text{y-hat}_{s,d}^{\text{persist}} = y_{s,d-1} \quad (2)$$

The same-day study compares feature ablations S, W, SW, SW+G, SW+G+T and SW+G+T+H. Ridge regression uses alpha = 1.0, fitted preprocessing and a scaled target. Random forest uses 150 trees, a minimum leaf size of 5 and a feature fraction of 0.5. The tabular XGBoost candidate uses 400 trees, depth 6, learning rate 0.05, row/column sampling of 0.8 and minimum child weight 3.

Selection uses the designated chronological selection month, not held-out-year scores. The current same-day history-enhanced model is Ridge-history. An environment-only Ridge comparator is retained to show how much performance depends on historical AQI rather than satellite and weather inputs alone.

C.2 Next-day GRU ensemble

The next-day model uses a gated recurrent unit (GRU) sequence encoder [2]. Each input window contains the 28 calendar days preceding the target. Every time step has six channels: historical AQI, an availability indicator, sine/cosine of that history day's annual position, latitude and longitude. The target-day AQI is never included in its input window.

$$X_{s,d} = [x_{s,d-28}, \ldots, x_{s,d-1}] \quad (3)$$

$$h_t = \text{GRU}(x_t, h_{t-1}); \quad \text{y-hat} = \text{Dense}_1(\text{ReLU}(\text{Dense}_{32}(\text{Dropout}(h_{28})))) \quad (4)$$

A single GRU layer has hidden size 64, followed by dropout 0.1, a 32-unit ReLU layer and a scalar output. Three independently trained seeds (0, 1 and 2) form the ensemble. Each seed's prediction is constrained to the reported AQI range 0-500, and the final estimate is their arithmetic mean.

Table 3 GRU training configuration.

Setting	Implemented value
Optimizer and objective	Adam; learning rate 0.002; weight decay 0.0001; mean squared error
Training	Batch size 256; at most 60 epochs; gradient norm clipping at 1
Early stopping	Patience 8; latest 15% of unique fitting dates used for internal validation
Boundary control	27-day purge for the 28-day window; scaling fitted on internal training data only

The chosen checkpoint is not refitted using the outer selection month, calibration block or test year. Hence the declared fitting period also contains the separate internal early-stopping block; not every row in that period is used for gradient updates.

C. Proposed Model / Algorithm (continued)

C.3 Weather-aware residual forecasting

The weather candidate uses XGBoost [3] to model the change from the latest eligible AQI rather than requiring a tree ensemble to learn the persistence level from scratch. Its features include 28 AQI lags, rolling summaries, the latest AQI change, target-day seasonality, station location and the five archived weather variables. Residual output is added to the previous-day anchor and constrained to 0-500.

$$r_{s,d} = y_{s,d} - y_{s,d-1}; \quad \text{y-hat}_{s,d} = \text{clip}(y_{s,d-1} + g(X_{s,d}), 0, 500) \quad (5)$$

The 2026 development grid contains four candidates: history-only and history-plus-forecast-weather models at depths 3 and 5. Each uses 600 trees, learning rate 0.035, subsample and column sampling of 0.85 and 0.9, L2 regularization 5, minimum child weight 15 and seed 2. The weather-aware depth-5 model is selected on October 2025 data. Its parameters and artifacts are frozen before 2026 target scoring.

C.4 Attention research models and controls

The dedicated research suite compares uniform aggregation, temporal attention (TA), feature attention (FA) and combined TA+FA. Sequence models use a GRU encoder with hidden size 64. Temporal attention scores hidden states through a 32-unit tanh scoring layer and softmax. Feature attention produces normalized feature weights and rescales the weighted inputs by the feature count. Uniform aggregation is a control for the gain attributable to learned attention.

$$a_t = \text{softmax}_t(v^T \tanh(W h_t + b)); \quad c = \text{sum}_t\, a_t\, h_t \quad (6)$$

Spatial variants compare uniform pooling, temporal attention, spatial attention and their combination. Environmental cell histories are encoded with shared recurrent parameters. Cell-presence masks exclude absent locations from aggregation. Relative offsets provide spatial context. These models support a distinct multimodal attention analysis; their attention outputs are not transferred to the plain forecasting models.

C.5 Fixed-model faithfulness audit

For a saved research checkpoint, attention rankings are compared with SHAP rankings [4] and feature-group replacement effects. The checkpoint is held fixed: this measures output sensitivity, not performance after retraining without a group. Deletion curves replace increasingly many ranked groups and record mean absolute prediction change. Structural metadata, including cell-presence and relative-offset information, is preserved during relevant spatial perturbations.

Replacement uses standardized mean values and training-season donor values in the implemented audit variants. Spearman rank agreement and city/season summaries are reported for the recorded audit sample and checkpoint version. Finite samples, correlated inputs and potentially unrealistic replacement combinations limit interpretation. Rank agreement alone does not establish that attention is a faithful or causal explanation.

D. Training & Testing Strategy

D.1 Chronological fitting, selection, calibration and testing

Future-year evaluation uses separate fitting, model-selection, interval-calibration and test blocks. Hyperparameters are chosen by selection-month RMSE. Calibration targets determine interval width only. Neither selection nor calibration is performed on the test year. Normalization, imputation, early stopping and donor backgrounds use only their permitted fitting subsets.

Table 4 Implemented outer chronological benchmark blocks. Counts are final benchmark cohorts, not raw export totals.

Benchmark	Fitting period	Selection / calibration	Held-out targets
2024 estimation	Jan-Sep 2023	Oct 2023 / Nov-Dec 2023	2024: 28,803 station-days
2025 forecast	Jan 2023-Sep 2024	Oct 2024 / Nov-Dec 2024	2025: 30,263 station-days
2026 weather candidate	Jan 2024-Sep 2025	Oct 2025 / Nov-Dec 2025	Jan-Sep 2026: 20,086 station-days

The 2026 comparison evaluates the frozen weather candidate, the previously frozen GRU ensemble and persistence on an identical paired cohort. All eight cities are represented, but only 91 of the configured 107 stations contribute eligible test targets. Export coverage is checked independently: 846 of 963 station-month exports were available across 94 stations, and three additional stations had exports without usable daily targets. Exclusion records distinguish unavailable exports from failed daily quality requirements.

D.2 Separate attention generalization protocols

The earlier attention suite includes chronological, leave-one-city-out (LOCO) and interleaved seasonal cross-validation experiments, with three neural seeds. LOCO tests spatial transfer and seasonal folds test held-out temporal regimes. Their fitting sets can contain dates later than some held-out observations; consequently these experiments are not presented as strictly prospective future-year forecasting. Only the chronology described in Table 4 supports that claim.

D.3 Computing environment and reproducibility

GPU training uses an NVIDIA RTX 4060 Laptop GPU with 8 GB memory. PyTorch runs the recurrent models through CUDA, and XGBoost uses GPU histogram training. The recorded benchmark environment includes PyTorch 2.10 with CUDA 12.8, XGBoost 3.4.1, NumPy 2.1.2, pandas 2.2.3 and scikit-learn 1.6.0. Saved models can be reloaded for CPU inference checks.

Each benchmark saves its input/source hashes, split definitions, seeds, parameters, checkpoints, predictions, metrics and station exclusions. The 2026 test protocol and candidate artifacts are sealed before scoring. An independent verifier reconstructs target dates and persistence references, checks shared-row metrics and interval bounds, validates hashes and reloads selected model predictions. Dashboard API responses and downloadable files are checked against the saved results.

Reproduction follows this sequence: acquire and validate exports; build station-day targets; join eligible features; construct calendar histories; fit candidates on the fitting block; select and freeze artifacts; calibrate intervals; score the paired test cohort; then verify and publish the historical dashboard outputs.

E. Evaluation Metrics

E.1 Prediction error and persistence improvement

RMSE is the primary selection metric, while MAE and R-squared describe absolute error and explained variation. Metrics are computed using paired observed and predicted values on eligible station-days. Let e_i = y-hat$_i$ - y_i, n be the number of evaluation rows, and y-bar be their observed mean.

$$\text{MAE} = (1/n)\ \text{sum}_i\ |e_i|; \quad \text{RMSE} = \text{sqrt}((1/n)\ \text{sum}_i\ e_i^2) \quad (7)$$

$$R^2 = 1 - [\text{sum}_i\ e_i^2 / \text{sum}_i\ (y_i - \text{y-bar})^2] \quad (8)$$

Overall and city-level metrics are reported, along with signed bias where available. Overall pooled metrics give greater weight to cities with more eligible station-days. City-chart metrics are separately calculated from paired daily city means; they are not averages of individual station RMSE values. City means weight available stations equally and do not represent the official published city AQI.

E.2 Paired uncertainty and calibrated prediction intervals

Model differences are assessed with 2,000 paired calendar-week block bootstrap resamples. All station observations within a sampled week remain together, and both models use the same sampled weeks. The 2.5th and 97.5th percentiles form a 95% confidence interval for the RMSE gain. This retains part of the temporal and shared-weather dependence while avoiding an independent-row bootstrap assumption.

$$\text{Gain} = \text{RMSE}_{\text{reference}} - \text{RMSE}_{\text{candidate}} \quad (9)$$

Nominal 90% intervals use absolute residuals from the dedicated calibration block. For n_{cal} residuals, the quantile level is $\min(1, \text{ceil}(0.9(n_{cal} + 1))/n_{cal})$, evaluated with the implementation's higher-quantile rule. The resulting radius q is applied symmetrically and constrained to the AQI range.

$$L_i = \max(0, \text{y-hat}_i - q); \quad U_i = \min(500, \text{y-hat}_i + q) \quad (10)$$

Observed interval coverage and width are reported rather than assuming a guarantee under temporal dependence or year-to-year distribution shift. These station intervals are not averaged and presented as calibrated uncertainty for city means.

E.3 Sharp-change detection and promotion rule

A sharp change is defined as an absolute AQI movement of at least 50 from the prior-day persistence reference. A correct detection requires a forecast movement of at least 50 in the same direction. Recall is correct detections divided by observed sharp changes; precision is correct detections divided by forecast-flagged changes. Zero denominators are treated explicitly rather than converted into apparently valid scores.

The predeclared candidate-promotion rule requires positive lower confidence bounds for RMSE improvement against both the frozen GRU and persistence, together with no loss of sharp-change recall or precision. This criterion is not revised after viewing test results. Attention faithfulness uses the separate agreement and perturbation metrics described in C.5.

E. Evaluation Metrics (continued)

E.4 Implemented experiment status

The following values identify the completed benchmark artifacts as of 10 October 2026. They document the current state of the methodology and are not additional criteria used to choose models after testing. Performance across different years should not be compared as though each model faced the same data distribution.

Table 5 Saved out-of-year benchmark results. AQI-point RMSE is rounded to three decimal places.

Held-out cohort	Model / comparator	RMSE
2024, 28,803 rows	Same-day Ridge-history / persistence	32.807 / 36.197
2024, same paired study	Environment-only Ridge	63.239
2025, 30,263 rows	Next-day GRU ensemble / persistence	32.516 / 35.388
2026, 20,086 shared rows	Weather-aware XGBoost	30.270
2026, same shared rows	Frozen GRU ensemble	31.707
2026, same shared rows	Persistence	34.668

On the 2026 shared cohort, the weather-aware model has MAE 19.345 and R-squared 0.848. Its paired RMSE gain over the frozen GRU is 1.437 AQI points, with a 95% weekly-bootstrap confidence interval of 0.376-2.614. The gain over persistence is 4.398, with interval 3.190-5.715. City-specific RMSE is improved in all eight represented cities.

Table 6 Sharp-change outcomes on the identical 2026 evaluation rows; threshold = 50 AQI points.

2026 sharp-change measure	Weather candidate	Frozen GRU
Observed sharp changes	2,315	2,315
Forecast-flagged changes	799	268
Correct directional detections	422	154
Recall	18.23%	6.65%
Precision	52.82%	57.46%

The weather candidate improves average error and sharp-change recall, but its precision falls. It therefore fails the predeclared promotion rule. The GRU remains the default next-day model, and the weather-aware candidate is available as a separately identified historical comparison. The weather model's nominal 90% interval has radius 61.166 and observed 2026 coverage of 94.88%; this retrospective coverage does not guarantee future calibration.

E.5 Dashboard presentation and study limitations

The dashboard provides prediction-mode and city selection, observed-versus-predicted curves, persistence comparisons, model and coverage summaries, station-level outputs and downloadable benchmark files. Forecast points are aligned to their target dates, and city plots use paired available stations. Historical evaluation output is distinguished from a newly issued forecast.

The main limitations are incomplete station exports, missing endpoint readings, unequal city coverage, coarse city-level forecast weather, satellite missingness, unverified real-time source latency and limited attention-audit samples. The observed weakness on sharp changes motivates additional forecast development, but any new candidate requires a newly declared validation protocol and independent evaluation before promotion.

References

[1] S. Jain and B. C. Wallace, "Attention is not Explanation," *NAACL-HLT*, pp. 3543-3556, 2019. DOI: 10.18653/v1/N19-1357.
https://aclanthology.org/N19-1357/

[2] K. Cho et al., "Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation," *EMNLP*, pp. 1724-1734, 2014. DOI: 10.3115/v1/D14-1179.
https://aclanthology.org/D14-1179/

[3] T. Chen and C. Guestrin, "XGBoost: A Scalable Tree Boosting System," *ACM SIGKDD*, 2016. DOI: 10.1145/2939672.2939785.
https://arxiv.org/abs/1603.02754

[4] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," *Advances in Neural Information Processing Systems*, 2017.
https://arxiv.org/abs/1705.07874

[5] NASA LP DAAC / Google Earth Engine, "MCD19A2.061: Terra & Aqua MAIAC Land Aerosol Optical Depth Daily 1km." Implemented band: Optical_Depth_047, scale 0.001.
https://developers.google.com/earth-engine/datasets/catalog/MODIS_061_MCD19A2_GRANULES

[6] ECMWF / Google Earth Engine, "ERA5-Land Daily Aggregated - ECMWF Climate Reanalysis." Collection: ECMWF/ERA5_LAND/DAILY_AGGR.
https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_DAILY_AGGR

[7] Open-Meteo, "Previous Runs API." Archived earlier-run forecast retrieval and previous_day1 semantics.
https://open-meteo.com/en/docs/previous-runs-api

Project evidence and data provenance

The implemented methods and numerical status tables were checked against the project's data-preparation, training, audit and verification modules, together with the saved 2024, 2025 and 2026 benchmark reports. The 2026 evidence package includes the frozen protocol, predictions, metrics, station-coverage and exclusion files, model/source manifests and independent verification report. These local artifacts are the source of experiment counts and outcomes; the publications above provide algorithmic context.

CPCB reported AQI exports and configured station metadata provide the ground-data provenance. Sentinel-5P OFFL L3 NO2 and HCHO are extracted through Google Earth Engine alongside the products in [5]-[6]. Data downloads, quality filters, feature availability and experiment hashes are retained in the project artifacts. Online documentation was consulted on 10 October 2026.